

SiC Power Device Research Scientist - Design

Wolfspeed · Durham, North Carolina

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Job description

At Wolfspeed, we do amazing things in a human way.

We know that the achievements of our organization are due to the passion, hard work and creativity of our employees. We celebrate different perspectives to foster excellence across our organization, and our goal is to make diversity a foundation of what we do. We are proudly building an environment where you can bring your authentic self to work.

- Enjoy doing things that people say can't be done? Innovation is at the center of everything we do.
- Hate red tape? We remove roadblocks instead of creating them.
- Working parent? We provide childcare assistance and paid parental leave.
- Student? We offer continuing education assistance.
- Looking for community? There are many ways to get involved, from Employee Resource Groups to local outreach.

Here's the Gist

In this position within Wolfspeed's Power Device Technology Development department, you will be leading device development projects for next-generation silicon carbide (SiC) power devices (MOSFETs, JFETs, JBS diodes and high voltage IGBTs) with an emphasis on power MOSFETs. You will be participating in all aspects of silicon carbide power device development, with a focus on device design efforts. You will interact with key customers as well as colleagues from Process Development, Test and Validation, Reliability, and New Product Introduction to master understanding of the full product life cycle of the new technologies.

The Day-to-Day:

- Lead device development efforts in high performance SiC power switching devices
- Participate in device concept development activities in various SiC device structures, including SiC power MOSFETs, JFETs, BJTs, IGBTs, PiN diodes, JBS diodes, and GTO thyristors
- Explore new device concepts for future generation of SiC power devices
- Perform Design of Experiment (DoE) activities for performance optimization of commercial SiC power devices
- Provide optimum device designs with considerations to manufacturability and reliability
- Hands on characterization experimental devices
- Data analysis of experimental data
- Interact with teams in power electronics applications, fab process development, reliability, and modeling, and deliver device designs with optimal characteristics for target applications.
- Prepare technical presentations and papers for internal meetings, as well as conferences and journals

This Job is Right for You if You Have (Requirements):

- Ph.D. in Electrical Engineering, Solid State Physics, or closely related area
- 5 - 10 years of industrial experience in power device development preferred
- Experience in low voltage discrete vertical silicon power MOSFET development preferred
- Strong understanding in semiconductor device physics
- Experience in hands-on characterization of power semiconductor devices
- Working knowledge of commercial TCAD tools
- Working knowledge of CAD layout tools
- Understanding of semiconductor fabrication processes and process integration
- Understanding of semiconductor device characterization methods
- Design of experiments and data interpretation skills
- Ability to work in a team setting
- Excellent analytical and problem-solving skills
- Excellent communications/presentations skills
- Must be authorized to work in the USA

This role may require additional duties and/or assignments as designated by management.

To put it legally - Wolfspeed is an equal opportunity employer.

We recruit, employ, train, compensate and promote regardless of race, sex, religion, color, national origin, disability, age, veteran status, gender identity, sexual orientation and other

protected status as required by law.

We are an equal opportunity employer and all qualified applicants will receive consideration for employment without regard to race, color, religion, sex, sexual orientation, national origin, disability status, protected veteran status, or any other characteristic protected by law.